

How Much of Truth Fits on a Single Axis?

Knowledge-Dependent Dimensionality, Polarity Recovery,
and the Anatomy of a Truth Direction in Qwen2.5-1.5B

Francesco Karim Vicidomini

Independent researcher

Reference code: `truth_probe.py`, `canvas.py`, `anatomy.py` and companion tools

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Abstract

Reading whether a language model represents a statement as true or false usually relies on trained linear probes or on sparse autoencoders. We study a minimal alternative: a single direction obtained, with no training, from the singular value decomposition (SVD) of hidden-state differences over true/false minimal pairs; the direction is identified without labels up to one global sign (Lemma 1). On Qwen2.5-1.5B, under a conservative held-out protocol, we report four findings. First, the axis carries a real truth signal (held-out AUC 0.938), yet a full linear probe reaches 0.989: one direction does not exhaust the representation. Second, the gap between axis and probe grows as the model’s knowledge of the facts decreases, and grows further when heterogeneous material forces one axis to serve many truth frames: the dimensionality of the truth representation is knowledge-dependent. Third, on mixed-polarity data the same decomposition recovers the supervised polarity direction of Bürger et al. (cosine 0.959) without seeing polarity labels. Fourth, attention builds the signal in the middle layers and the feed-forward blocks erode it after the peak, a claim supported causally by ablation. Seven falsification experiments designed to reveal richer structure fail to overturn the one-dimensional reading. Results are from a single small model and are intended to be falsifiable.

1 Introduction

Large language models produce false statements with the same fluency as true ones. Predicting when a model is about to produce a false statement, ideally during generation and without a second judge model, is a central reliability problem. Mechanistic interpretability has shown that many high-level properties, including the truth value of a statement, are encoded as linear directions in the residual stream [2, 4].

Two families of methods have been developed to read the truth value of a statement from hidden states. The first is the *linear probe*: a classifier or difference-of-means axis whose projection separates true from false [1, 2, 3]. The second is *sparse dictionary learning*: project the state onto an overcomplete dictionary, potentially millions of atoms, to undo superposition and read monosemantic features [6, 7].

Each pays a price. A fixed axis reads a single linear functional of the state, and prior work documents a concrete failure mode of such probes under negation [5]; a sparse autoencoder requires

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dedicated training and a large matrix multiply per token. This paper takes a third stance. Rather than learning a probe or expanding a dictionary, we read structure from geometry alone, and ask a precise question: *how much of the truth structure does a single direction capture, and when does that capture fail?* We extract all hidden states in one forward pass and let an axis emerge from the SVD of paired true/false differences. Part of the apparent accuracy of any such pipeline could come from selecting the best layer after the fact; we therefore measure everything out of sample and quantify the selection effect directly with a permutation null (Section 4.4). The answer is modest but, we believe, clean: the axis reads truth cheaply when the fact is known, fails predictably when it is not, and, blind to the label, recovers the polarity structure that a prior supervised method constructed by hand.

A single positive result invites the obvious objection: is the one-dimensionality real, or merely the easiest thing to see? The second half of the paper responds with a systematic robustness analysis. We subject the one-dimensional claim to seven falsification experiments, each designed to expose a richer structure if one exists, and report that none overturns it. We then go one level deeper and ask *where* the axis comes from: decomposing the residual stream into its attention and feed-forward (FFN) contributions, we locate the components that build, carry, and erode the signal, and test the resulting picture causally by ablation.

Contributions.

1. A training-free truth probe at cost $O(d)$, where d is the residual-stream width, identified without labels up to one global sign (Lemma 1), with a held-out and permutation-tested evaluation that corrects the optimism of in-sample axis selection (Sec. 5.1).
2. Evidence that the *dimensionality of the truth representation is knowledge-dependent*: near one-dimensional for known facts, diffuse for unknown ones, quantified by the gap to a full linear probe (Sec. 5.2).
3. *Recovery of the polarity direction t_P* of Bürger et al. [5] without polarity supervision: the SVD of mixed-polarity differences aligns with the supervised t_P at cosine 0.959 (Sec. 5.4).
4. A *robustness battery*: seven falsification experiments against the one-dimensional reading (a second dimension, depth-global structure, dynamics, layer departments, energy, phase, three-dimensional clustering), each tested under held-out cross-validation and a permutation null, none of which overturns the claim (Sec. 6).
5. An *anatomy* of the axis: an exact attention/FFN decomposition of the residual stream with a built-in correctness gate, showing a stable per-layer alternation in readability, a negative result inside the expanded FFN space, and causal ablations establishing that attention builds the signal in the middle layers while the FFN degrades it in the transition band (Sec. 7).
6. An interpretation of the residual overlap on unknown facts as the observable signature of unresolved superposition, tied to a capacity argument (Secs. 8, 10).

2 Related work

Linear truth directions. [1] extract a knowledge direction without labels by enforcing logical consistency across a statement and its negation; the direction is found by optimization. [2] show truth has emergent linear structure and that difference-of-means probes generalize as well as more

complex ones. [3] intervene along truthful directions at inference time; [4] frame representation reading top-down. Our axis shares the linear-direction hypothesis, but it is read from the geometry of paired differences by an SVD, with no optimization; its precise relation to supervision is stated in Lemma 1.

A two-dimensional truth subspace. Closest to us, Bürger et al. [5] show that truth is represented not in a single direction but in a two-dimensional subspace spanned by a general truth direction t_G and a polarity-sensitive direction $t_P \sim \text{XOR}(\text{is_true}, \text{is_negated})$, and that probes trained on affirmative statements fail on negated ones. We confirm this structure in Qwen2.5-1.5B and show, as a methodological addition, that t_P can be recovered *without* polarity supervision.

Cross-layer dynamics. Recent monitoring methods model the evolution of hidden states across depth rather than reading a single layer [10, 11]. Our depth trajectories are descriptive and consistent with this line.

Beyond single-direction probes on the residual stream. All the methods above, ours included, assume white-box access to the residual stream, which is not always available and is not the only usable signal. Bar-Shalom et al. [9] treat the full activation tensor (layers by tokens) as an image and train a vision-transformer detector on it, across several LLMs simultaneously, with evaluation on a broad set of hallucination benchmarks. Methods of this kind are more expressive than any single direction and are trained for detection; our axis sits at the opposite end of the same cost/expressivity curve, as the cheapest structural point, and the comparison we draw is against that curve, not against any single method.

Superposition and dictionaries. Superposition [8] motivates decomposing the residual stream into overcomplete dictionaries [6, 7]. This is the operation we explicitly do *not* perform. The bottom line, argued in Secs. 8 and 10, is that the two approaches answer different questions and neither subsumes the other: a dictionary answers “which features are active” at cost $O(dm)$ and after dedicated training; the axis answers “how much truth a single direction carries” at cost $O(d)$ with no training. When feature identity matters, dictionaries are the right tool; for a cheap runtime signal on facts the model knows, the axis suffices. We make no claim of superior accuracy over dictionary methods.

3 Scope and evidence

Everything in this paper concerns one small base model, Qwen2.5-1.5B, and sentence-level factual statements. Results come in two evidential tiers, and we keep them distinct throughout. **Tier 1** (Secs. 5 and 6): the positive findings and the falsification battery, evaluated held-out with permutation nulls. **Tier 2** (Sec. 7): the anatomy, a mechanistic follow-up with the same protocol but smaller margins in places, single datasets for some entries, and no falsification battery of its own; where a statement there is interpretation rather than measurement, it is marked as such. Cross-model generalization is not claimed here; what is known about it from subsequent work is summarized, with numbers, in Section 11.

4 Method

4.1 Extraction

Informally: a transformer reads a sentence and, at every layer, updates an internal vector summary of the text so far; we record that summary at the position of the last token, where the sentence is complete, at every layer, and study the geometry of these vectors.

Formally, we load Qwen2.5-1.5B frozen. The model has 28 transformer blocks, hence 29 hidden-state levels (the embedding plus each block output), each a vector of width $d = 1536$, the width of the residual stream. Per sentence we run one forward pass and take the last-token state at every level. For N sentences this yields $H \in \mathbb{R}^{N \times 29 \times d}$. Once an axis is fixed, scoring a new state is a single dot product, hence the reading cost of $O(d)$ per token quoted throughout.

4.2 Data: minimal pairs

A *minimal pair* is a pair of sentences with the same subject and the same template that differ in exactly one content word, such that one sentence is true and the other false: “The Sun is a star.” vs. “The Sun is a planet.” Both sentences are affirmative. Within a pair, the difference of hidden states cancels the shared topic and surface form and leaves the truth signal.

Two sources are used. The *curated pairs* are a built-in set of 45 pairs of common facts (capitals, chemical symbols, authorship, currencies) that the model demonstrably reproduces in open-ended completion (Sec. 12); they play the role of a known-facts control: behaviorally verified and token-matched, hence also clean, in the sense of free of the sentence-format confounds discussed in Sec. 14. The external datasets are CounterFact [12] and TruthfulQA [13] (250 pairs, pinned revisions in Sec. 13), which contain many relations a 1.5B model plausibly does not know.

To quantify this, we define the *know-rate*: the fraction of facts the model reproduces correctly in open-ended greedy completion. Over 200 CounterFact prompts, Qwen2.5-1.5B has a know-rate of 30.0% (61/200), which grounds the “plausibly does not know” reading; the know-rates of the other models appear in Sec. 11.

Why a flipped predicate and not a negation. The false counterpart is built by flipping a content word (“planet” for “star”), not by inserting “not”. The choice is deliberate, for three reasons. First, negation is not a neutral edit: it activates a dedicated polarity direction in the representation [5], so a negated counterpart would entangle the truth signal with the polarity signal. Our design keeps both sentences affirmative and isolates truth; polarity is then studied separately and deliberately in the 2×2 design of Sec. 4.6. Second, inserting “not” changes the token count and shifts all subsequent positions, weakening the token-level matching that makes the intra-pair difference cancel shared structure. Third, the two designs are complementary rather than competing: the negated forms are exactly the NT/NF cells of the polarity design, so the alternative is not ignored but treated as its own object of study.

4.3 The axis, and where labels enter

We form the intra-pair difference matrix D with rows $h_{\text{true}} - h_{\text{false}}$ at a given layer, and take its SVD,

$$D = USV^\top, \quad v_1 = V_1, \tag{1}$$

where v_1 , the top right-singular vector, is the dominant direction of the differences. The role of supervision in this construction can be stated exactly.

Lemma 1 (Label-free identification of the axis, up to sign). *Let each pair i contribute a row $d_i = s_i(h_i^A - h_i^B)$ to D , where h_i^A, h_i^B are the two hidden states of the pair and $s_i \in \{\pm 1\}$ encodes which sentence of the pair is designated true. Then $D^\top D = \sum_i d_i d_i^\top$ does not depend on (s_i) ; hence the right singular vectors of D , and in particular the span of v_1 , are invariant to the within-pair labels.*

Proof. $d_i d_i^\top = (-d_i)(-d_i)^\top$ for every i , so each summand of the Gram matrix, and therefore the Gram matrix and its eigenvectors, is unchanged under any sign assignment. $\square$

Labels enter the pipeline at exactly two points. First, a single global orientation bit (“the true side is positive”), fit on training folds, fixes the sign of v_1 . Second, the evaluation itself: AUC is defined with respect to labels. Everything upstream of these two points, the pairing, the differences, and the axis, uses no labels. We therefore describe the axis as *training-free* (no optimization of any kind) and *label-free up to one sign bit* (Lemma 1), and we reserve the word “unsupervised” for the polarity recovery of Sec. 5.4, where the recovered structure (polarity) is never labeled at all. The lemma also explains, in advance, the behavior of the permutation null below: permuting which sentence of a pair is called true leaves the span of the axis unchanged and destroys only the orientation and the test labels, which is why the null concentrates near chance.

4.4 A falsifiable evaluation protocol

The central methodological point is that the axis, its calibration, the orientation bit, *and* the choice of peak layer (the layer whose held-out AUC is maximal, chosen on training folds) must be fit on training data only. We use k -fold cross-validation *over pairs*: a pair is never split across folds, which would leak the shared topic. Everything is fit on the training folds and we report held-out AUC averaged across folds. Pairs from the same CounterFact relation template may still fall in different folds; generalization across topics is therefore not certified by this protocol, and we return to the point in Sec. 14.

Part of the apparent accuracy of any layer-scanned pipeline is selection: the best of 29 layers looks good by chance. The *permutation null* quantifies exactly that contribution. Let A_{obs} be the observed maximum held-out AUC over layers (“AUC is the area under the ROC curve, equivalently the probability that a randomly chosen true statement receives a higher score than a randomly chosen false one, with ties counted as one half.”). For $b = 1, \dots, B$ we randomly swap true/false *within* each pair (the labels become noise while the pair structure is preserved), rerun the identical pipeline including the layer scan, and record the maximum held-out AUC A_b . The estimated p -value is the standard conservative estimator

$$\hat{p} = \frac{1 + \#\{b : A_b \geq A_{\text{obs}}\}}{B + 1}. \quad (2)$$

With $B = 200$ and no null value reaching the observed statistic, $\hat{p} = 1/201 \approx 0.005$, the value reported in Table 1. As a reference point we also fit an ℓ_2 -regularized logistic probe on the raw d -dimensional states, held out over the same pair folds.

4.5 A complex-plane extension

We additionally form a second axis v_2 , the SVD residual orthogonal to v_1 , treat the projection as $z = c_1 + i c_2$ with calibrated real and imaginary parts, and test whether magnitude- or phase-based scores improve separation over the real axis alone. They do not (Sec. 5.3); the negative finding is kept because it is itself informative.

Table 1: Held-out AUC (5-fold CV over pairs) at layer 16, Qwen2.5-1.5B. “1D” is the real axis; “MAG” the magnitude-gated polar score; “PHASE” the angular-deviation score; “2D” a logistic on (Re, Im); “Probe” an ℓ_2 logistic on raw states. The last column is the label-permutation $\hat{p}$ for the 1D axis.

Dataset	pairs	1D	MAG	PHASE	2D	Probe	$\hat{p}$
Curated pairs	45	0.938	0.936	0.928	0.931	0.989	0.005
CounterFact only	250	0.792	0.790	0.771	0.795	0.854	0.005
CounterFact / TruthfulQA mix	250	0.638	0.640	0.676	0.651	0.841	0.005

4.6 Polarity design

To test whether a genuine second dimension exists, we use a 2×2 design: each fact in four forms, affirmative-true (AT), affirmative-false (AF), negated-true (NT), negated-false (NF), giving the XOR structure of [5]. We compute the affirmative and negated truth directions $t_A = \overline{AT} - \overline{AF}$, $t_N = \overline{NT} - \overline{NF}$, the general direction $t_G \propto t_A + t_N$, and the supervised polarity direction $t_P \propto t_A - t_N$. For the recovery test, we run the difference SVD on minimal pairs that *mix* the two polarities, so the SVD never sees the polarity label, and compare its leading directions to the supervised t_P .

4.7 Residual decomposition and its correctness gate

Within a block the residual stream is updated additively,

$$h_{\ell+1} = h_{\ell} + a_{\ell} + f_{\ell}, \quad (3)$$

where a_{ℓ} is the attention contribution and f_{ℓ} the FFN contribution, both captured by forward hooks. The residual stream studied everywhere else in this paper is the sum; the anatomy of Sec. 7 reads the two addends separately.

Every anatomy tool embeds an **identity check**: it verifies that $a_{\ell} + f_{\ell}$ reconstructs the residual delta $h_{\ell+1} - h_{\ell}$. The verdict uses the *median* relative error across (sentence, layer) rather than the maximum: where a block adds almost nothing the delta is near zero and the ratio explodes, so the maximum is fragile while the median reports the typical case. Observed median error: about 1.0×10^{-7} on both datasets, so the decomposition is exact and the anatomy numbers are trustworthy. (The single inflated layer, block 27, has a near-zero residual delta and is excluded from interpretation.) Anatomy runs use `float32`: the identity fails spuriously in `bfloat16` because subtracting large residual states to form small per-block deltas suffers catastrophic cancellation.

5 Results I: signal, dimensionality, polarity

All numbers are on Qwen2.5-1.5B with last-token pooling; AUC is out-of-fold unless stated.

5.1 A real but bounded truth signal

Table 1 reports the held-out evaluation at the peak layer (layer 16). On the curated pairs the training-free axis reaches AUC 0.938 with permutation $\hat{p} = 0.005$ (null mean 0.588, 95th percentile 0.664): the geometric axis aligns with truth well beyond what layer selection alone can manufacture. The signal is therefore real. It is also *bounded*: a full linear probe on the same raw states reaches 0.989. The probe sees information the single axis discards.

5.2 The gap scales with model knowledge, and with heterogeneity

The rows of Table 1 differ not only in difficulty but in the *size of the probe gap*, and the three rows separate two distinct effects.

The first effect is knowledge. On curated facts the model demonstrably knows, the axis (0.938) trails the full probe (0.989) by 0.051. On CounterFact alone, where the measured know-rate is 30.0% (Sec. 4.2) and many relations are obscure (for example, the mother tongue of a minor historical figure), the separability of the axis falls to 0.792 and the gap widens to 0.062 (probe 0.854). Less knowledge, lower separability, and a modestly larger gap.

The second effect is heterogeneity. On the CounterFact/TruthfulQA mix the gap jumps to 0.203 (axis 0.638, probe 0.841), far more than either ingredient alone would suggest. A single direction fit across heterogeneous material must serve all of it at once, while the full probe is free to combine directions; the excess gap on the mix therefore measures how much the truth direction itself varies with the material. Subsequent work confirms this reading independently (Sec. 11, item 4): the per-category truth axes form a structured, semantically signed geometry, and forcing one axis onto a mixture of categories is exactly what the mix row pays for.

We read the knowledge effect as a statement about *dimensionality*. When the model knows a fact, its truth content is concentrated and a single direction captures most of it. When the model does not know a fact, the truth-relevant signal is weaker and more diffuse, and one direction recovers less of it. The dimensionality of the truth representation is thus not fixed but *knowledge-dependent*; the mix row adds the complementary statement that it is also *category-dependent*. This joint framing is, to our knowledge, stated in these terms by neither the linear-probe nor the dictionary-learning literature, and it is the contribution we consider most novel.

5.3 The complex plane does not help; “phase” is the sign flip

On single-polarity data the second dimension adds nothing: at layer 16 the 2D logistic (0.931) does not exceed the 1D axis (0.938) on curated pairs, and on CounterFact it gains +0.003 (0.795 vs. 0.792). This is structural. Minimal pairs cancel the topic, so the differences concentrate along one direction; the SVD places the entire signal in v_1 , and v_2 is residual variance, noise rather than a second truth factor. A controlled synthetic check confirms that the apparent “rotation” of falsehoods is an artifact: with a pure sign flip and zero genuine rotation, an angular (phase) score still separates classes, because a state at $\text{Re} < 0$, $\text{Im} \approx 0$ has phase $\approx \pi$ by definition. The earlier intuition of truth-as-phase has no operational content here: it reduces to the sign of the real axis re-expressed in polar coordinates. We retract that framing.

5.4 Recovery of the polarity direction without polarity labels

A genuine second dimension does exist, but only when the data span both polarities, and it is the polarity direction of [5], not a phase. Table 2 shows the diagnostic: an axis fit on affirmative statements classifies affirmative held-out facts at 0.981 but *anti-classifies* negated ones at 0.079, a near-total flip, while the general direction t_G recovers the negated facts at 0.954. The truth direction genuinely rotates with polarity ($\cos(t_A, t_N) = -0.415$).

The methodological result is the recovery. When we run the difference SVD on *mixed-polarity* minimal pairs, blind to the negation label, its leading direction aligns with the supervised polarity direction t_P at cosine 0.959 (layer 15), and t_P lands in the first principal component, while alignment with t_G is only 0.157. In other words, when both polarities are present, the dominant axis of the differences *is* polarity, and the SVD finds it without supervision; by Lemma 1 the recovery cannot

Table 2: Polarity analysis (leave-facts-out CV), layer 14 chosen on affirmative data; and unsupervised t_P recovery at layer 15. AUC below 0.5 indicates a sign flip.

Quantity	Value
$\cos(t_A, t_N)$	-0.415
Affirmative axis on affirmative (held-out AUC)	0.981
Affirmative axis on <i>negated</i> (held-out AUC)	0.079
General direction t_G on negated (held-out AUC)	0.954
SVD vs. supervised t_P : $\max_k \cos(\text{SVD}_k, t_P) $	0.959
(principal component carrying t_P)	PC 1
$\cos(\text{SVD}_1, t_G)$	0.157

have used the labels even implicitly. Where [5] constructs t_P by separating affirmative from negated statements by hand, the same direction emerges here from geometry alone.

6 Robustness: seven falsification experiments

The first half of this paper established a positive claim. We now stress-test it. The claim under test is:

In Qwen2.5-1.5B, the truth value of a statement is carried by a single (one-dimensional) direction, the position axis obtained from the SVD of paired true/false differences, on facts the model knows; the second and higher dimensions add no separation.

A claim of one-dimensionality is falsifiable in a specific way: exhibit a second, third, or depth-distributed coordinate that separates true from false better than the single axis. We construct seven such coordinates and test each.

Two kinds of test, kept distinct. We separate *controlled experiments* from *visual observations*, and label every result as one or the other. Controlled experiments (experiments 1–4) produce a held-out AUC under k -fold cross-validation over pairs and a label-permutation null that quantifies selection bias; a result is meaningful only if it clears that null. Visual observations (experiments 5–7) do not classify; they project the states and measure a single descriptive statistic, the per-coordinate separation AUC or the centroid-separation ratio, so the conclusion follows from a number attached to a picture, not from a fitted model. Throughout, the model is frozen Qwen2.5-1.5B, last-token pooling, peak layer 16 unless stated.

Table 3 summarizes all seven with their real numbers; the prose below gives the reasoning for each.

1. The complex plane. The original framing promoted the second SVD direction to an imaginary axis and read truth as a phase rotation. If real, a phase- or magnitude-based score should beat the real axis. It does not: on curated pairs the phase score reaches AUC 0.928 against 0.938 for the real axis, and a controlled synthetic check shows that with a pure sign flip and *zero* genuine rotation an angular score still separates, because a state at $\text{Re} < 0$ has phase $\approx \pi$ by definition. The “rotation” is the sign of the real axis re-expressed in polar coordinates. The complex framing is decorative; we retract it. (This is the negative result of Sec. 5.3, restated as the first falsification experiment.)

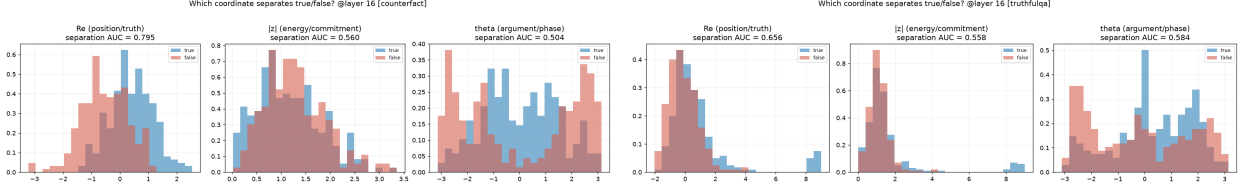
Table 3: Seven experiments attempting to falsify the one-dimensional truth axis. “Exp.” = controlled experiment (held-out CV + permutation null); “Obs.” = visual observation (descriptive separation statistic). None overturns the claim.

#	Richer structure hypothesized	Type	Result (real numbers)	Verdict
1	A complex/phase second dimension (truth as rotation)	Exp.	PHASE AUC 0.928 vs. 1D 0.938 (curated); reduces to the real-axis sign flip	falsified
2	Truth is global across depth (fixed-axis signature)	Exp.	global 0.923 vs. local 0.938 (−0.015), $\hat{p} = 0.0099$; persistence gap +0.04	falsified
3	Truth is dynamic, a “butterfly effect” in the unsaturated shade, on hard data	Exp.	global 0.642 vs. local 0.627 (+0.015), $\hat{p} = 0.0099$; commitment true 2.83 $\rightarrow$ 5.91 vs. false 2.65 $\rightarrow$ 5.86	falsified
4	Truth is distributed across layer “departments”	Exp.	diagnostic curve is a smooth ramp (no steps); global 0.646 vs. local 0.627 (+0.019), $\hat{p} = 0.0099$	falsified
5	The argument θ (phase) carries truth	Obs.	θ AUC 0.504 / 0.584 / 0.556 (CF / TQA / mix)	falsified
6	The energy $ z $ (commitment) carries truth	Obs.	$ z $ AUC 0.560 / 0.558 / 0.543	falsified
7	True/false form distinct 3D clumps	Obs.	centroid ratio 0.55 / 0.51 / 0.48 (< 1 , overlapping); Re alone 0.795 / 0.656	falsified

2. A global depth signature. If truth flows through the network, the full trajectory $[\Delta_0, \dots, \Delta_{28}]$ should beat the single best layer, where the signed deviation at each layer ℓ is defined as $\Delta_\ell = b_\ell - \frac{1}{2}$, with $b_\ell = \sigma(\text{Re}(z_\ell))$ representing the calibrated projection passed through a sigmoid along a fixed axis. On curated pairs it does not: the depth-signature classifier reaches held-out AUC 0.923 against 0.938 for the single layer (−0.015), and although the signature is real (permutation $\hat{p} = 0.0099$), it carries no advantage. The crossing-persistence gap between false and true is +0.04, flat (where for each sentence, the first layer at which the trajectory crosses below zero and the fraction of subsequent layers it stays below; the gap is the difference of this persistence between false and true statements). No propagation.

3. A dynamic butterfly effect in the shade. The signed deviation saturates (σ compresses large values to $\pm\frac{1}{2}$), so a propagating *uncertainty* could be invisible to it. We re-ran on hard data (CounterFact/TruthfulQA mix) with the unsaturated features $[\text{Re}, |z|]$, the “shade”. The depth signature now beats the single layer, but by +0.015 (0.642 vs. 0.627, $\hat{p} = 0.0099$): real and marginal. The decisive descriptive measure kills the butterfly effect: the commitment magnitude grows identically for both classes (true 2.83 $\rightarrow$ 5.91, false 2.65 $\rightarrow$ 5.86); false statements do not hesitate more than true ones, and there is no propagating divergence.

4. Layer departments. If different layer bands perform different jobs (surface, semantics, logic), a single semantic axis re-read everywhere wastes most layers; giving each layer its *own* native probe should recover a distributed signal, and the per-layer separation curve should show steps. It shows a smooth ramp instead: the per-layer native separation rises continuously to a single peak at layer 16,



(a) CounterFact: Re 0.795, $|z|$ 0.560, θ 0.504

(b) TruthfulQA: Re 0.656, $|z|$ 0.558, θ 0.584

Figure 1: Experiments 5–6 and knowledge scaling, layer 16. Per-coordinate separation AUC. On both datasets only the position axis (Re) separates true from false; energy ($|z|$) and argument (θ) are at chance. The position separation falls from 0.795 (CounterFact, often-known facts) to 0.656 (TruthfulQA, more ambiguous), while the one-dimensional structure is unchanged.

with no plateaus and no jumps. The per-layer signature beats the single layer by +0.019 (0.646 vs. 0.627, $\hat{p} = 0.0099$), again real and marginal. No departments; the native-axis “cure” improves the fixed axis but does not change the ceiling.

5–6. Energy and argument. Decomposing each state into position (Re), energy ($|z|$), and argument (θ) and measuring how well each separates true from false (Fig. 1): only position carries signal. The energy separates at $|z|$ AUC 0.560/0.558/0.543 (CounterFact/TruthfulQA/mix): true and false are equally committed. The argument separates at θ AUC 0.504/0.584/0.556, essentially chance; what little it shows is the echo of the Re sign, not independent information. The phase, the very coordinate the complex framing made central, is empty.

7. Three-dimensional clumping. If truth were a richer object, true and false should form distinct clusters in the three-dimensional (v_1, v_2, v_3) space. They do not (Fig. 2). The centroid-separation ratio, the distance between the true and false centroids divided by the average cloud spread, is 0.55/0.51/0.48, all below 1: the clouds overlap. Crucially this low ratio is *not* evidence against the axis; it is the one-dimensional signal seen through two noise dimensions. Projected onto position alone, the same data separate at AUC 0.795 (CounterFact). Adding the orthogonal dimensions dilutes, it does not enrich.

6.1 What survives

Three statements resist all seven experiments.

Truth is one-dimensional and lives on the position axis. Across every decomposition, only Re separates; energy, phase, the second and third SVD directions, and the full-depth trajectory add at most +0.02 AUC and usually nothing.

Separability scales with knowledge. On CounterFact, where the model often knows the fact, the position axis separates at AUC 0.795; on the more ambiguous TruthfulQA at 0.656; on the diffuse mix the centroids nearly coincide. The clearer the model’s knowledge, the more one-dimensional and separable the truth representation.

On unknown facts, the classes show low separability. Where the model does not know the fact, true and false representations do not form distinct clusters but overlap in space, intuitively a dense, uniform distribution rather than two distinct clumps. We state this as *low class separability / distribution overlap* (centroid ratio < 1), the measured counterpart of the knowledge-dependent dimensionality of Sec. 5.2.

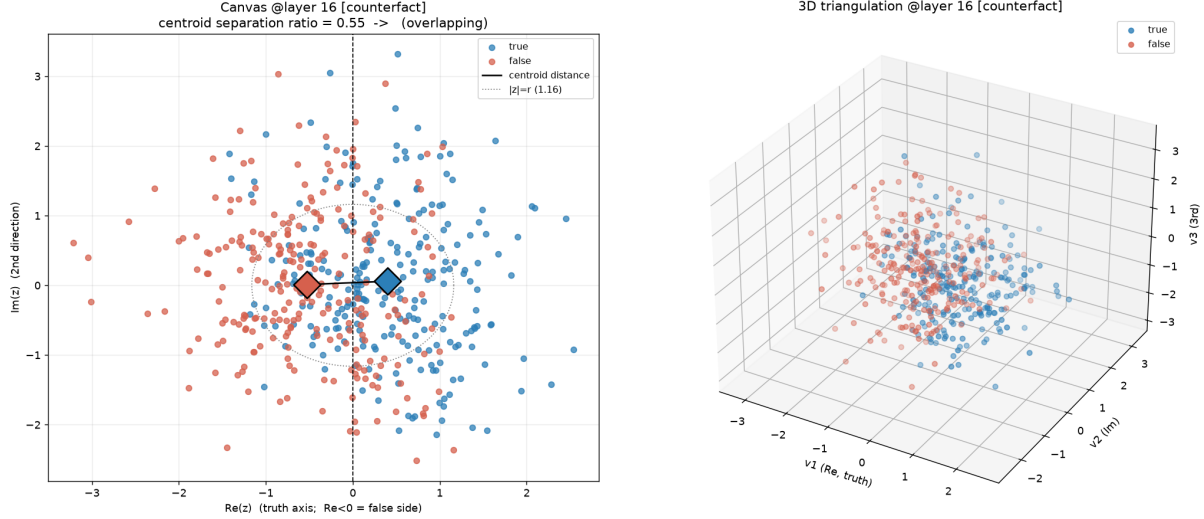


Figure 2: Experiment 7, CounterFact at layer 16. Left: the 2D plane; the true (blue) and false (red) centroids (diamonds) are close relative to the cloud spread (centroid-separation ratio 0.55, overlapping). Right: the three-dimensional (v_1, v_2, v_3) scatter shows a gradient along v_1 (position) but no distinct clumps. The separation is confined to the horizontal position axis; the extra dimensions are noise.

7 Results II: the anatomy of the axis

The falsification battery established *that* truth is one-dimensional on known facts. This section asks *how the residual stream comes to carry it*: which architectural component, attention or the FFN, produces, holds, or degrades the signal, layer by layer. The results here are Tier 2 in the sense of Sec. 3: same protocol and shields, but smaller margins in places, and interpretations are marked as such. The decomposition and its correctness gate are defined in Sec. 4.7.

7.1 Per-layer readability: a stable alternation

For each layer we fit the truth axis on a_ℓ and on f_ℓ separately and measure held-out AUC, repeating over 5 cross-validation seeds. A per-layer difference $\text{AUC}(a_\ell) - \text{AUC}(f_\ell)$ is counted as real only if its magnitude exceeds both 2σ (its own across-seed standard deviation) and a floor of 0.03. Table 4 shows the peak block.

Table 4: Truth AUC by component at the readability peak (block 15), CounterFact, mean $\pm$ std over 5 seeds. Residual = the summed stream (reference).

	residual	attention	FFN
AUC	0.792 ± 0.003	0.845 ± 0.003	0.734 ± 0.004

The alternation survives re-seeding and reproduces structurally across both datasets. On CounterFact, attention stably dominates readability at layers $\{6, 7, 10, 12, 13, 14, 15, 17, 18\}$ and the FFN at $\{8, 11, 16, 19, 20, 21, 22, 23, 25, 26\}$. At the peak (block 15), where the truth axis is read, attention leads by $+0.110 \pm 0.004$. **Interpretation (not yet causal):** attention appears to carry the truth signal most strongly through the middle band, including the peak. Readability, however,

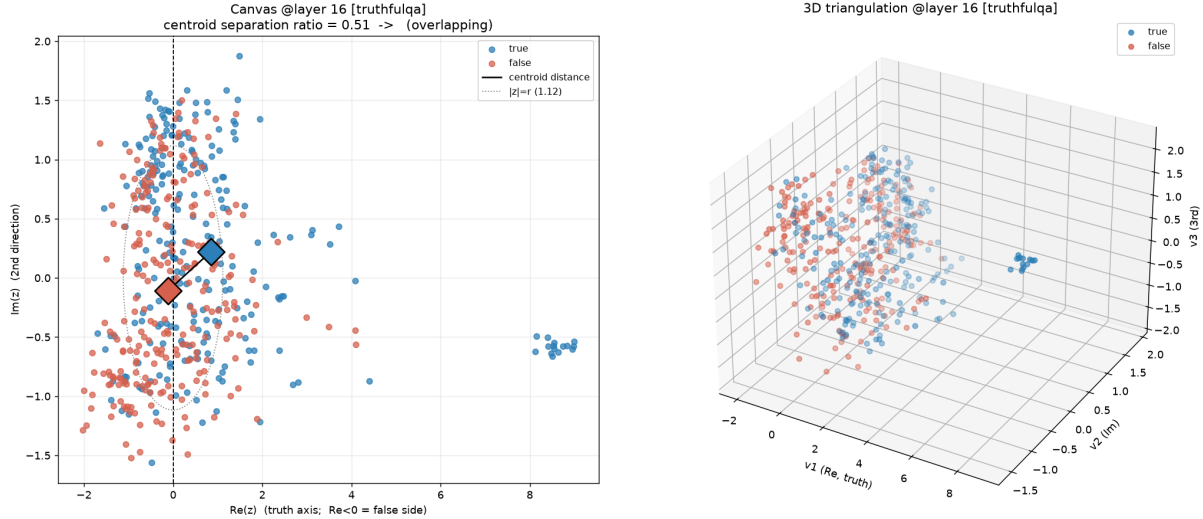


Figure 3: Experiment 7, TruthfulQA at layer 16. The same picture on more ambiguous facts: centroid-separation ratio 0.51 (overlapping), no three-dimensional clumps. Lower knowledge, lower separability, identical one-dimensional structure.

is not necessity, which Sec. 7.3 tests directly, and which changes the picture.

7.2 The expanded FFN space does not help

The FFN expands its input to 8960 dimensions, applies SwiGLU, and re-compresses it. One might expect the truth signal to be more cleanly readable inside that larger, putatively sparser space, before compression. We capture the expanded activation (the input to `down_proj`) and fit the truth axis there. In this $p \gg n$ regime the permutation null is elevated, so only the margin over the null counts: at the peak the expanded axis clears its null cleanly (curated 0.953 vs. null 95th percentile 0.669, margin +0.284, $\hat{p} = 0.0099$; CounterFact 0.816 vs. 0.561, margin +0.254, $\hat{p} = 0.0099$). **But relative to the compressed residual at the same layer, the expanded space gains essentially nothing:** +0.017 (curated) and +0.023 (CounterFact). Raw architectural expansion does not undo superposition on these facts. The implication for future work is concrete: sparsity must be *imposed*, for example by a trained sparse autoencoder, not merely afforded by a wider space.

7.3 Causal ablation: attention builds, the FFN erodes

To separate readability from necessity we zero a component’s contribution over a band of layers (a hook returns zeros, so $h + 0 = h$), let the effect propagate, and measure the collapse of truth AUC at a downstream readout (Table 5).

Table 5: Causal ablation on CounterFact (250 pairs). Drop = intact – ablated; positive drop = the component is causally necessary.

band (layers)	readout	intact	attention-off (drop)	FFN-off (drop)
11–15 (middle)	block 15	0.793	0.487 (+0.306)	0.819 (−0.026)
16–18 (transition)	block 18	0.712	0.708 (+0.004)	0.815 (−0.103)
19–26 (late)	block 27	0.537	0.534 (+0.003)	0.539 (−0.002)

Three readings, in order of confidence:

- **Middle band: attention is causally necessary.** Removing attention collapses truth from 0.793 to 0.487 (drop +0.306); removing the FFN does nothing (−0.026). Attention does not merely read the signal here, it builds it.
- **Transition band: the FFN *degrades* the signal.** Removing attention again does nothing (+0.004), but removing the FFN *raises* truth AUC from 0.712 to 0.815 (drop −0.103). From layer 16 onward the FFN injects content that lowers true/false separability. This is the first causal support for an earlier conjecture, that the FFN adds context which amplifies the overlap on uncertain facts, though it rests on one band and one dataset and should be treated as a lead, not a result.
- **Late band: nothing remains to destroy.** The intact AUC is already about 0.54 (near chance): the truth signal has evaporated toward the output, so neither ablation has an effect.

This falsifies the initial “attention builds, FFN consolidates” hypothesis. There is no late consolidation; the FFN’s apparent late *readability* dominance (Sec. 7.1) was the readability of a dying signal, not active maintenance. The corrected picture: attention constructs the truth representation in the middle layers; the FFN begins to erode it in the transition band; by the late layers it is gone, as the model shifts from representing the fact to predicting the next token.

8 Discussion: what a single axis can and cannot read

It is tempting to view a cheap geometric probe as a low-cost version of dictionary learning, reading superposition without expanding it. We argue this is impossible, and that the impossibility is a capacity statement, not an implementation gap. A projection is an inner product: it collapses a d -dimensional state to a single scalar, discarding $d - 1$ dimensions. From that scalar one cannot reconstruct which features were present, only how much a chosen direction was. “Reading the superposition” requires asking the question once per dictionary atom, which over an overcomplete dictionary of size m is exactly the $O(dm)$ sparse autoencoder. Compression (project to a scalar) and disambiguation (expand to a sparse code) are not inverse cheap-vs-expensive versions of one operation; they are different operations, and only the second reads superposition. The gaps of Sec. 5.2, 0.062 on homogeneous unknown material and 0.203 on the heterogeneous mix, are direct measurements of the truth information that a single axis structurally cannot return.

The consequence is a clear scope. This method’s value is *structural and cheap*: a training-free axis that, on facts the model knows, rivals a trained probe at two dot products per token; a quantitative law relating the readability of truth to the model’s knowledge of the fact; and a route to the polarity direction that needs no polarity labels. It is not a substitute for sparse autoencoders and should not be presented as one.

9 The axis reads truth and confidence

The seven experiments establish that there is no second coordinate of truth to find. This has a constructive consequence that is easy to miss, so we state it plainly. The same axis admits *two* readings, not one.

Its *sign* classifies a statement: $\text{Re} > 0$ is the true side, $\text{Re} < 0$ the false side. Its *separability*, how cleanly true and false split on a given set of facts, measures how well the model knows that material. The geometry that answers “is this statement true?” therefore also answers, indirectly, “does the

model know this?”, because a model that does not know produces overlapping rather than separated representations (Figs. 2, 3). These are not two axes; they are two readings of one axis: the direction of the arrow, and the sharpness of the arrow.

The reading is quantitative at the level of a distribution. The position-axis separation is 0.795 on CounterFact and 0.656 on TruthfulQA: by this measure the model “knows” the CounterFact material better than the TruthfulQA material, read off the geometry alone, with no external grading. We report this as a property of the dataset, not yet of the single fact. Calibrating a per-fact confidence, a local separability around one statement that predicts whether the model actually errs on it, is the natural next step, and requires a behavioral label (does the model answer the open question correctly?) that the minimal-pair design does not contain. We mark it as future work rather than a claim. Section 12 returns to this label and to a prerequisite it exposed.

10 Why unknown facts overlap: the signature of superposition

It remains to interpret *why* the classes overlap on unknown facts rather than simply asking *that* they do. The overlap is not measurement noise; it is the observable signature of unresolved superposition, and the explanation follows from the capacity argument of Sec. 8.

A projection onto a single direction returns a scalar: it collapses the d -dimensional state to one number and discards $d - 1$ dimensions. When the model knows a fact, its truth content has been allocated a clean, dominant direction of its own (the residual stream has, in effect, dedicated an axis to it), and the single projection recovers almost all of it (position AUC 0.795, near the full-probe ceiling). When the model does not know a fact, there is no dedicated direction: the weak truth-relevant signal remains distributed across many superposed features, and no single axis can pull it out of the tangle. Recovering it would require asking the question once per dictionary atom, the expansion that a sparse autoencoder performs at cost $O(dm)$, which is a different operation from a single projection, not a cheaper version of it.

The consequence is exactly what the visual observations show. On known facts the truth occupies its own direction, the classes form two regions, and the centroid ratio is high. On unknown facts the truth is spread across the superposition, the classes overlap, and the centroid ratio falls below one. The dense, uniform overlap of true and false on unknown facts is therefore not a failure of the probe but a direct, visual readout of superposition that the probe structurally cannot resolve. The probe gaps of Sec. 5.2 and the low centroid ratio measured here are views of the same quantity: the truth information that lives off-axis on facts the model does not know (0.062), inflated further when heterogeneous material forces one axis to serve many truth frames at once (0.203). Subsequent work (Sec. 11, item 4) begins to resolve that superposed remainder into named, signed components: what this paper can only call unresolved superposition acquires, there, a geometry.

11 Scope and generalization: what subsequent work established

Everything above concerns Qwen2.5-1.5B. The natural question, whether the same conclusions hold on other models, has been addressed in subsequent work by the author, available in the same repository, and we state the outcomes here as precise claims so the reader knows what is established and where.

1. **The signal and the peak structure replicate on four models across two families** (Qwen2.5-1.5B, Qwen2.5-3B, Llama-3.2-1B, Llama-3.2-3B): on each model the training-free axis clears its permutation null held-out, with a single readability peak at intermediate relative depth.

2. **The knowledge-gap relation gains cross-family support, with a family offset.** With knowledge measured behaviorally and uniformly (the know-rate of Sec. 4.2: 200 CounterFact prompts, greedy decoding, seed 0) and all gaps measured on CounterFact under the same held-out protocol, the four models give Table 6. Within each family the gap decreases as the know-rate increases (Qwen: 0.062 at 30.0% vs. 0.049 at 37.0%; Llama: 0.141 at 35.5% vs. 0.112 at 41.0%). Across families a family-level offset remains: at comparable knowledge, the Qwen models concentrate more of the truth signal on the single axis than the Llama models. The knowledge dependence of Sec. 5.2 is therefore not an artifact of one model, and the family offset is consistent with the family-trait finding of item 3.

Table 6: Behavioral know-rate (200 CounterFact prompts, greedy decoding, seed 0) and axis-vs-probe gap (CounterFact, 250 pairs, held-out), per model. The Qwen2.5-1.5B row is measured directly in this paper (Table 1); the other rows come from the companion work under the same protocol.

Model	Know-rate (correct/total)	Axis-vs-probe gap
Llama-3.2-3B	41.0% (82/200)	0.112
Qwen2.5-3B	37.0% (74/200)	0.049
Llama-3.2-1B	35.5% (71/200)	0.141
Qwen2.5-1.5B	30.0% (61/200)	0.062

3. **The anatomy generalizes as a relational law.** An intermediate result (that the FFN writes pro-truth into the peak layer and flips to anti-truth at exactly the layer after the peak) was published by the author and then corrected within days by an axis-provenance control built against the author’s own instrument; the correction is reported as a result. The corrected law, confirmed by pre-registered predictions on all four models: the FFN’s write at a block opposes every truth frame that does not contain it, while attention aligns with existing frames it did not write. At the peak, on the same block and the same frame, the two contributions are near mirror images (+1.63 for attention vs. −1.43 for the FFN on Llama-3.2-3B). The erosion reported in Sec. 7.3 survives the correction and is causally attributed, by exact algebraic split and by freezing interventions, to the SwiGLU value stream rather than the gate.
4. **The axis is a semantically signed mixture.** Fitting the axis per fact category shows that per-category truth axes form a structured geometry (related categories cluster with positive cosines, one category anti-aligns) and that this arrangement converges across the two model families (Mantel test, a permutation test for the correlation between two distance matrices, applied to the category-geometry matrices of the two families: $r = +0.77$, $p = 0.0009$). Within-category held-out AUC exceeds cross-category AUC, which bears directly on the topic-generalization question raised by the protocol of Sec. 4.4.

These four items carry their own held-out protocols, permutation nulls, and pre-registered predictions, and are documented, with reproduction commands, in the same repository as this paper. They are summarized here for scope, not claimed as results of this paper. Read in sequence, the chain is linear: the present paper isolates the axis and measures its limits; the follow-up work explains those limits, first mechanically (which component builds the axis and which erodes it, generalized as a relational law) and then geometrically (what the axis is a mixture of). The overlap that Sec. 10 can only call unresolved superposition is, by the end of that chain, partially resolved into named, signed components.

12 A note for future work: verify behavior before geometry

Two natural extensions, a per-fact behavioral confidence label (Sec. 9) and a study of the geometry of *logical* (as opposed to semantic) truth, share a prerequisite that is easy to overlook, and we record it here as methodological guidance rather than as a result.

An informal behavioral check (not a controlled experiment). Before building any geometric study of a capability, one should confirm that the capability exists behaviorally. We ran a small, qualitative probe of the base model in open-ended completion (greedy decoding, no context carried between prompts). Two things were apparent on a handful of prompts. First, the model reproduces the common semantic facts used throughout this paper (capitals, chemical symbols, authorship, currencies), consistent with the “facts the model knows” assumption behind the curated pairs. Second, on symbolic transitive syllogisms built from nonce words (“a *grok* is heavier than a *blarn*; a *blarn* is heavier than a *fendu*; therefore ...”), the base 1.5B model does *not* perform the deduction: it deflects into multiple-choice or code scaffolding rather than completing the transitive conclusion. We stress that this is an informal observation on a few prompts, not a measured result, and we draw no quantitative claim from it.

The principle. Studying the geometry of a capability presupposes the capability. Measuring the hidden-state geometry of a task the model cannot perform yields noise, and that noise is easily mistaken for “the property is distributed/complex” when the truthful reading is “the property is absent.” A cheap behavioral feasibility check should therefore precede any geometric study of *logical* truth; on this model that study is not feasible, which is itself useful to know in advance. The same principle applies to transfer: before studying the geometry of truth in a second model, one should first verify behaviorally that the second model knows the facts in question. (In the subsequent multi-model work of Sec. 11 this principle is applied systematically: the behavioral know-rate is measured before any geometry.)

A hypothesis to test, not a claim. The way the base model deflects on uncertain input, falling back on memorized *format* scaffolding (exam layouts, question/answer templates, code), suggests, but does not establish, that part of the overlap on unknown facts may carry a sentence-format component rather than pure superposition noise. This is consistent with the format confound that makes mixing datasets inadvisable (Sec. 14). We flag it as a direction for a future, properly shielded experiment, not as evidence.

13 Reproducibility

Every number in this paper is reproduced by one command. `truth_probe.py` runs the controlled experiments (the positive results and falsification experiments 1–4); `canvas.py` renders the visual observations (experiments 5–7); `anatomy.py`, `anatomy_consolidate.py`, `anatomy_expanded.py`, and `ablation.py` produce the anatomy of Sec. 7 and embed the identity, null, and stability shields described there. Each result in `truth_probe.py` is an *isolated* subcommand: running one task loads, extracts, and computes only its own data, so a reviewer who wants to verify one figure runs exactly one command. All scripts load the model with safetensors only and load external datasets as Parquet only, never with remote code. A CUDA GPU is optional.

All scripts are archived on Zenodo (DOI 10.5281/zenodo.20938285) and mirrored at github.com/Francesco-Marhel/TruthProbe.

Setup.

```
pip install torch transformers datasets
```

Positive results (Tables 1, 2).

```
# Table 1, top row (curated pairs): held-out AUC, ablation,  
# full-probe baseline, permutation p-value  
python truth_probe.py signal --dataset builtin --with-2d --baseline --perm 200
```

```
# Table 1, bottom row (CounterFact / TruthfulQA mix)  
python truth_probe.py signal --dataset mix --max-pairs 250 \  
--with-2d --baseline --perm 200
```

```
# Table 2, top (affirmative -> negated flip; the 2D polarity structure)  
python truth_probe.py polarity
```

```
# Table 2, bottom (SVD recovers the polarity direction t_P)  
python truth_probe.py recovery
```

Know-rate (Secs. 4.2 and 11) and the CounterFact row of Table 1.

```
python behav_check.py --model Qwen/Qwen2.5-1.5B --n 200 --seed 0
```

```
python truth_probe.py signal --dataset counterfact --max-pairs 250 --baseline \  
--perm 200 --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e
```

The same two commands with `-model` set to `Qwen/Qwen2.5-3B`, `meta-llama/Llama-3.2-1B`, or `meta-llama/Llama-3.2-3B` reproduce the corresponding rows of Table 6.

Controlled falsification experiments (Table 3, 1–4).

```
# experiment 1 (PHASE ablation) is the PHASE column of ‘signal’ above
```

```
# experiment 2: global depth signature (fixed axis), curated pairs  
python truth_probe.py domino --dataset builtin --perm 100
```

```
# experiment 3: dynamic shade on hard data  
python truth_probe.py domino --dataset mix --max-pairs 250 --signal-mag --perm 100 \  
--rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e \  
--rev-truthfulqa 741b8276f2d1982aa3d5b832d3ee81ed3b896490
```

```
# experiment 4: per-layer native axes + diagnostic ramp  
python truth_probe.py domino --dataset mix --max-pairs 250 --per-layer-axis --perm 100 \  
--rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e \  
--rev-truthfulqa 741b8276f2d1982aa3d5b832d3ee81ed3b896490
```

Visual observations (Figs. 1–3, experiments 5–7).

```
python canvas.py --dataset counterfact --layer 16 --out counterfact \
    --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e
python canvas.py --dataset truthfulqa --layer 16 --out truthfulqa \
    --rev-truthfulqa 741b8276f2d1982aa3d5b832d3ee81ed3b896490
```

Each `canvas.py` run writes `<out>_coords.png`, `<out>_plane.png`, and `<out>_3d.png`. Run CounterFact and TruthfulQA *separately*; do not pass `-dataset mix` to `canvas.py`, as mixing injects a spurious sentence-format axis.

Anatomy (Tables 4–5).

```
# per-layer attention/FFN readability (identity check embedded)
python anatomy.py --dataset counterfact --max-pairs 250 --perm 100 \
    --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e

# stability of the alternation across 5 seeds
python anatomy_consolidate.py --dataset counterfact --max-pairs 250 --seeds 5 \
    --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e

# the truth axis inside the expanded FFN space (8960-dim)
python anatomy_expanded.py --dataset counterfact --max-pairs 250 --perm 100 \
    --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e

# causal ablation, the three bands of Table 5
python ablation.py --dataset counterfact --max-pairs 250 \
    --band-start 11 --band-end 15 --readout 15 \
    --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e
python ablation.py --dataset counterfact --max-pairs 250 \
    --band-start 16 --band-end 18 --readout 18 \
    --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e
python ablation.py --dataset counterfact --max-pairs 250 \
    --band-start 19 --band-end 26 --readout 27 \
    --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e
```

Anatomy runs use `float32` (Sec. 4.7); about 6 GB of VRAM is enough for Qwen2.5-1.5B.

Data provenance and safety. The curated pairs and the 2×2 polarity set are embedded in `truth_probe.py`. The external datasets are NeelNanda/`counterfact-tracing` (commit `c945b08...`) and `truthful_qa`, *generation* config (commit `741b827...`), loaded as Parquet only. For bit-exact reproduction the dataset commits may be pinned as shown, or a local file passed with `-file-counterfact <path>`. Add `-h` to any subcommand for its options.

14 Limitations

These results are confined to a single, small model (Qwen2.5-1.5B); the entire battery, the positive results and all seven falsification experiments, should be re-run at larger scale before the one-dimensionality is treated as general (the partial replications available so far are summarized, with

their scope, in Sec. 11). It is plausible that departments, dynamics, or a genuine second dimension emerge with scale, precisely because a 1.5B residual stream may be too compact to separate them. The datasets are small (tens to a few hundred pairs), so confidence intervals are wide despite the permutation control. Cross-validation is over pairs, not over topics: pairs from the same CounterFact relation may fall in different folds, so generalization to entirely unseen topics is not certified by the protocol of Sec. 4.4; the within- vs. cross-category transfer measurements of subsequent work (Sec. 11, item 4) address the question directly. The “unknown-fact” interpretation of the CounterFact gap and of the low separability is plausible but confounded: a fact the model does not know and a fact whose truth is genuinely multidimensional are not separated here, and the low-separability reading is further confounded with sentence-format effects, which is why mixing datasets must be avoided. Disentangling ignorance from genuine multidimensionality (for example by filtering CounterFact to facts the model can answer, using the behavioral label of Sec. 12) is necessary to make the dimensionality claim quantitative. The anatomy results are Tier 2: margins in the expanded-space and transition-band analyses are modest and based on single datasets, the factual-knowledge label is noisy, and ablation measures necessity for *linear readability* of truth, not for the model’s behavior. We use last-token pooling; other aggregations may shift the axis.

15 Conclusion

Stripped of a decorative complex-plane framing, the positive results survive on a real model under a falsifiable protocol: a training-free truth axis, identified without labels up to one global sign, carries a permutation-tested signal at cost $O(d)$; the gap between that axis and a full probe scales with how well the model knows the fact, suggesting a knowledge-dependent dimensionality of truth; and the difference SVD recovers the supervised polarity direction of [5] at cosine 0.959 without polarity labels. The one-dimensional reading withstands seven falsification experiments, spanning a second dimension, depth-global structure, dynamics, departments, energy, phase, and three-dimensional clustering, none of which exhibits a richer separating coordinate. Beneath the axis, an exact decomposition of the residual stream locates its origin: attention builds the signal in the middle layers, causally, while the FFN erodes it after the peak, and by the late layers the model has moved on from representing the fact to predicting the next token. We read a result that resists seven falsifications as strengthened, not weakened, interpret the residual overlap on unknown facts as the visual signature of unresolved superposition, and regard the axis as a light, honest, almost-free structural probe to sit alongside generation. Natural next steps, partly realized in the subsequent work summarized in Sec. 11, are the multi-model study of the dimensionality law, a knowledge-controlled dataset built from a behavioral label, and the category-level structure of the axis, each preceded by a behavioral feasibility check of the kind described in Sec. 12.

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